Rebalancing the kinetochore

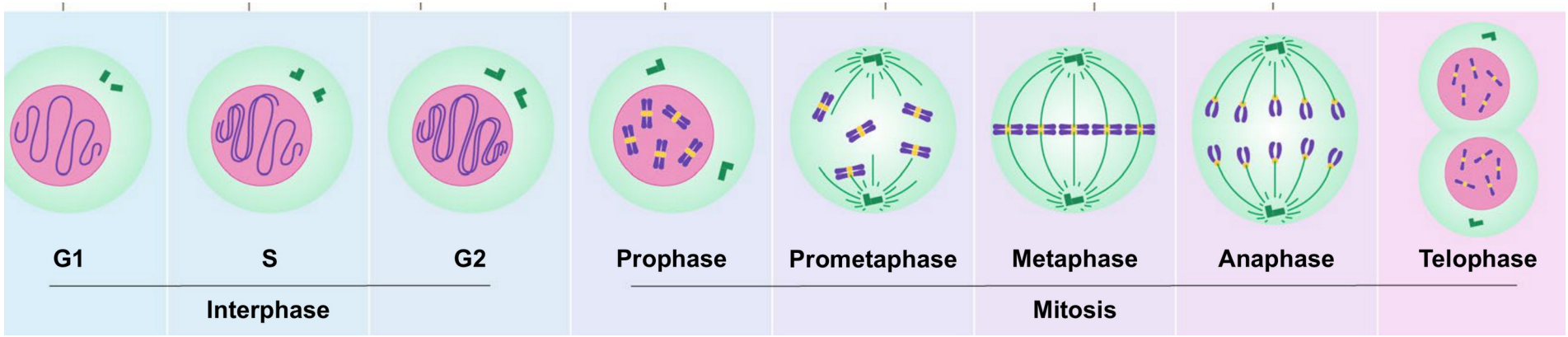
A drug-repurposing case for one child with mosaic variegated aneuploidy

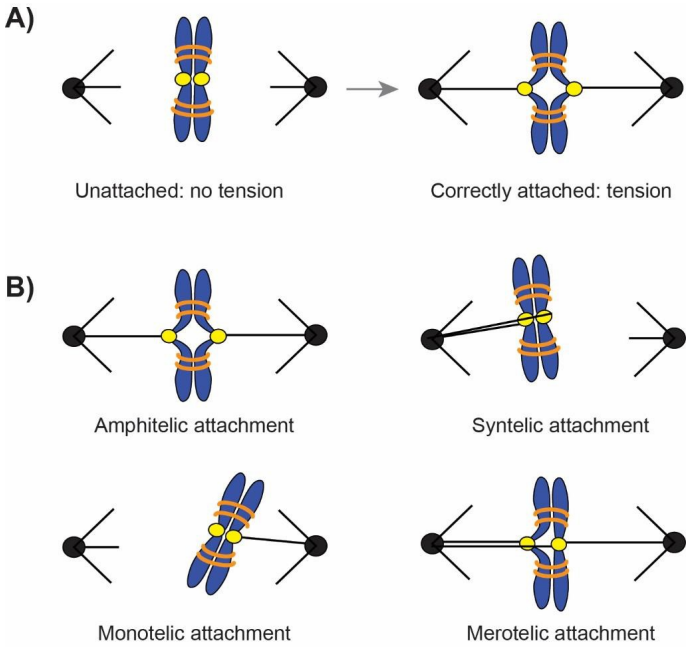
“Every cell-cycle drug in clinical use was built to break the cell cycle. Not one was built to make it work better.”

Maaz Asher Khurram · mk7790@nyu.edu

MVA Hackathon 2026 · Track 2 · proband WGS_EX2312012 · BUB1B p.Leu737Ter / p.Asn1002Lys

THE PROBLEM: every cell division is a chance to get the chromosome count wrong





A checkpoint guards the hand-off

Each chromosome must grip both poles. Three of the four ways it can attach are wrong, so the cell holds mitosis until every one is right.

In this child the guard is weak

MVA: cells are born with the wrong chromosome count, giving growth failure, microcephaly and childhood cancer risk.

Bloom & North 2021 PMID 33892776 · McVey 2021 PMID 34445523, CC BY 4.0

THE QUESTIONS we made this work answer, before we let ourselves pick anything

1 What do these two alleles actually leave behind?

Not “pathogenic”. How much protein, and with which domains.

2 Is the deficit reversible at all?

If restoring BubR1 does nothing, no drug can work. Someone tested this.

3 Is there any drug that raises BubR1, or its phosphatase?

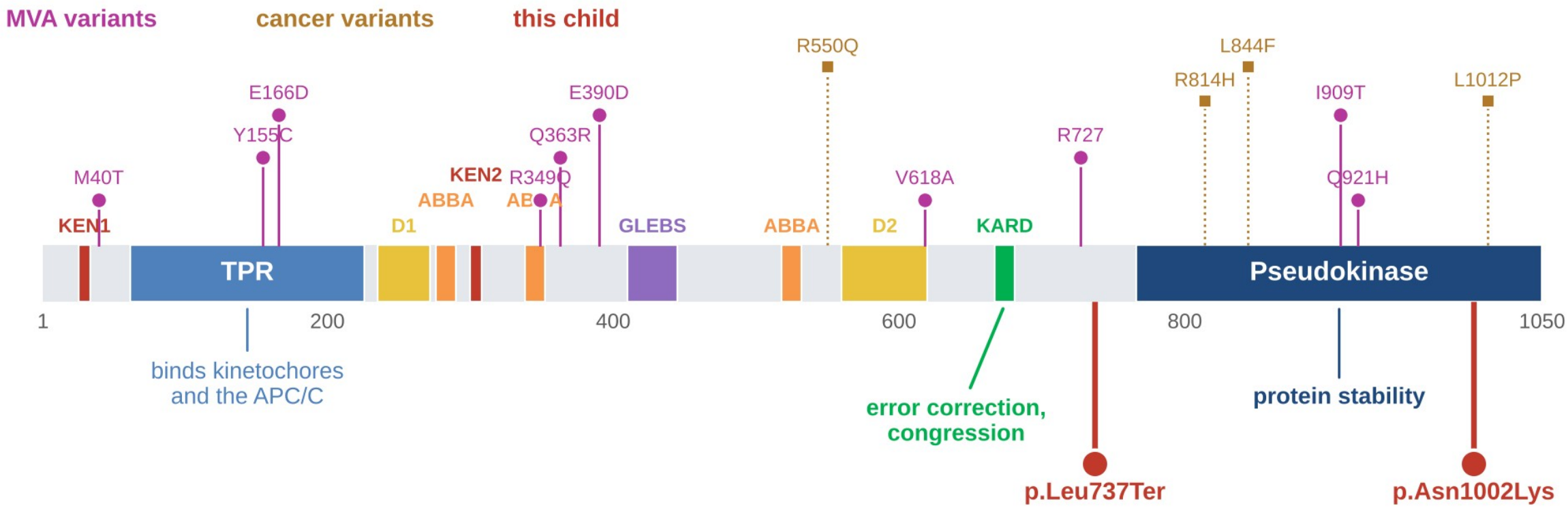
Asked across the whole druggable universe, not a shortlist.

4 If not, what else sits on the same node — and at what dose?

Answered on measured human exposure, or not answered.

Each question had to close before the next one opened

Q1 — Two alleles, and the domain that sits between them



Allele 1 truncates at 737

747 nt upstream of the last exon-exon junction, far past the 50-55 nt rule, so the transcript is destroyed.

Allele 2 is buried, not surface

87% buried. PREDICTED to be made, then degraded — a dosage mutant, not an interface one.

Both land on the KARD

667–681: the motif that recruits the phosphatase PP2A-B56.

Domain layout adapted from Bloom & North 2021, PMID 33892776

SCALABILITY (15%) — we scored every BUB1B allele, not this child's two

Scalability 15%

Every allele, scored the same way

2,536 alleles
atlas + 854 full texts

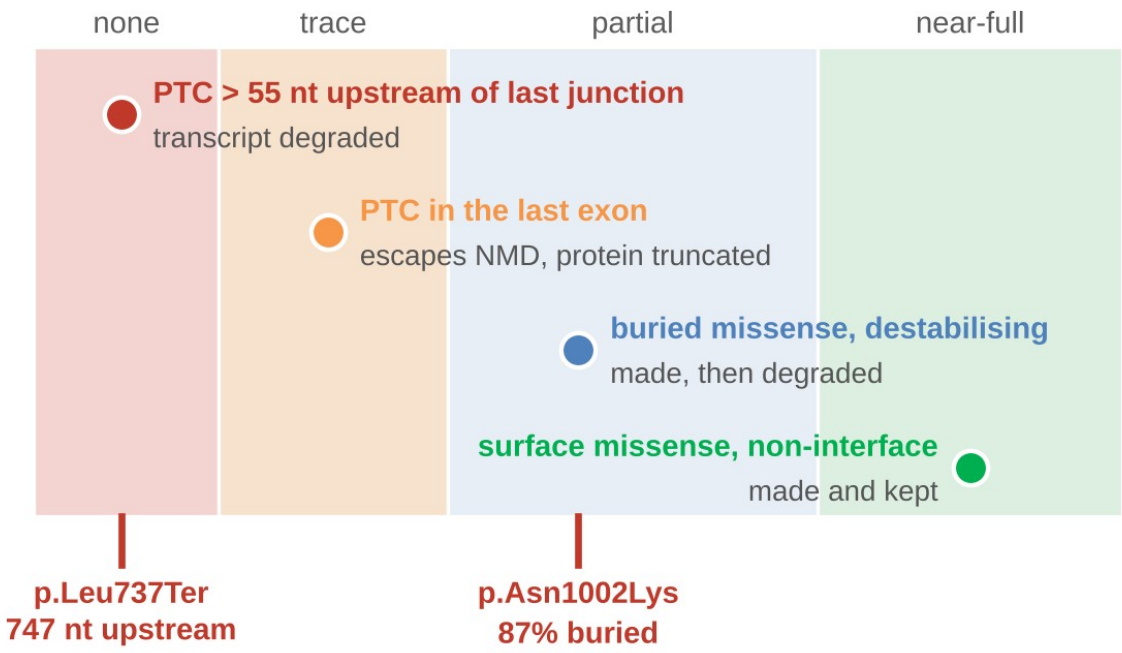
consequence call
truncating / missense / splice

NMD rule, 55 nt
does the transcript survive

AlphaMissense + GPN-Star
two predictors, one adds splice

literature abundance
no predictor has this axis

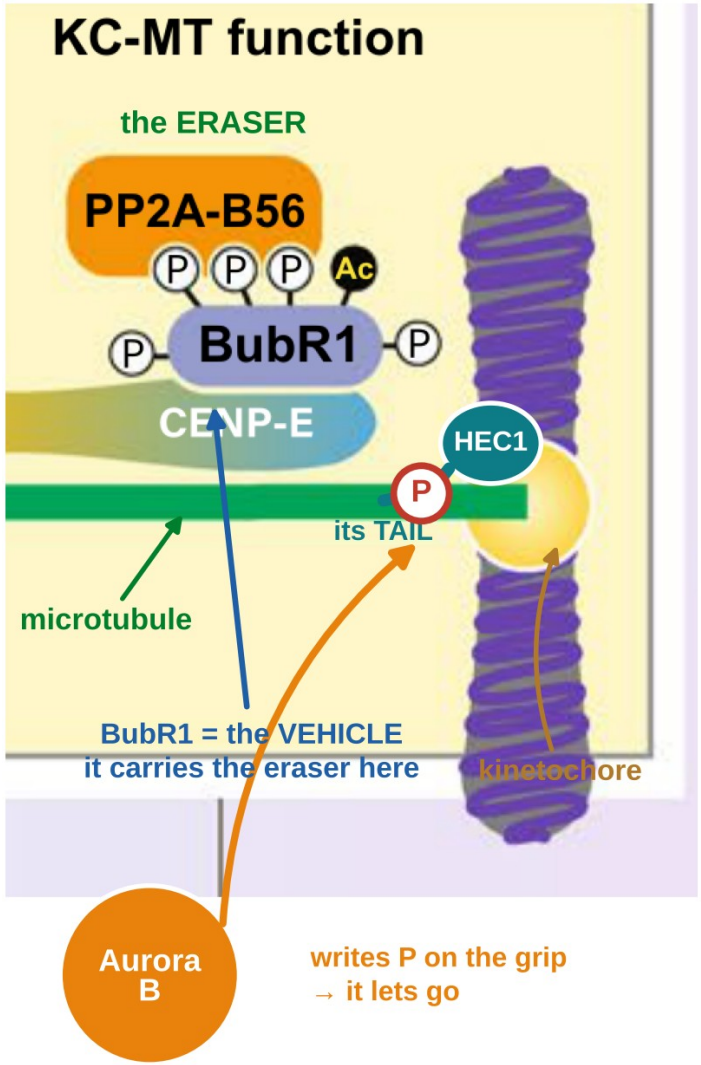
Expected BubR1 output, by allele class



Suijkerbuijk reached the same two classes experimentally: truncating alleles lose the transcript, missense alleles are made and turned over.

Suijkerbuijk et al., Cancer Res 2010, PMID 20516114

Q1 — Aurora B LOOSENS grips on purpose. PP2A-B56 SETS them. This child cannot set.



AURORA B — the loosener

Phosphorylates the Hec1 grip. A phosphorylated grip lets go — that is how wrong attachments get broken.

PP2A-B56 — the setter

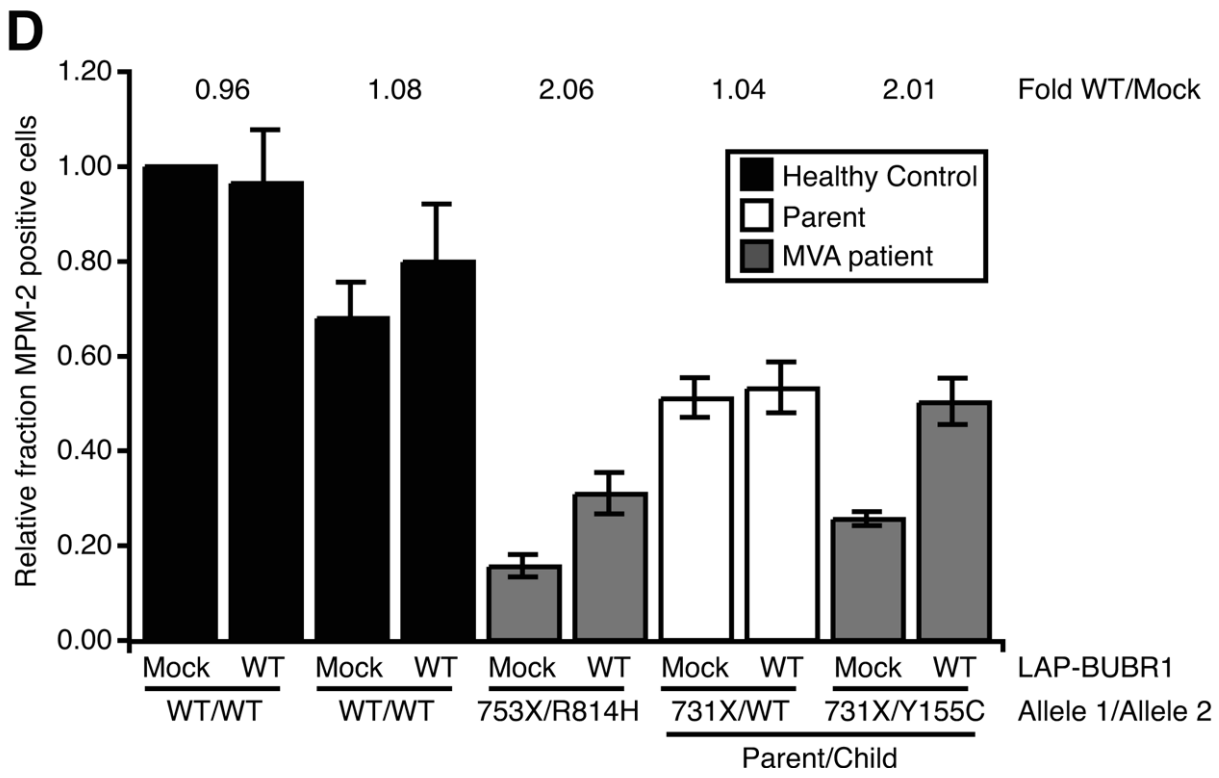
Removes those phosphates so a correct grip can set. Both are needed: this is a tug-of-war.

BubR1 carries the eraser. Plk1's 3 phosphates on its KARD are the TICKET PP2A-B56 binds — a different set from Aurora B's, opposite job.

No KARD delivered → no setter arrives → Aurora B is not stronger, it is UNOPPOSED. So we lower it to match.

Adapted from Bloom & North 2021, PMID 33892776, Fig 2

Q2 — The deficit is reversible: put BubR1 back and the checkpoint recovers



This child's genotype shape

753X/R814H and 731X/Y155C: one truncating allele, one missense — the MVA pattern.

2.06x and 2.01x

Nocodazole arrest doubles when BubR1 is restored. Healthy controls and the carrier parent do not move.

But it does not normalise

15% → 31% and 26% → 50% of healthy. And no drug raises BubR1.

Suijkerbuijk et al., Cancer Res 2010, PMID 20516114, Fig 1D

SCIENTIFIC RIGOR (35%) — Q3: ten arms, four themes, every one closed on a number

Scientific rigor 35%

Rescue allele 1
the truncated one

- ~~Translational readthrough~~
- ~~NMD inhibition~~
- ~~Gene / ASO / trans-splicing~~

closed on exposure

Rescue allele 2
the missense one

- ~~Chaperone, HSP90~~
- ~~Degradation, SIRT2 / NAD+~~
- ~~Proteasome inhibition~~

closed on exposure

Target BubR1 itself
make it, or bind it

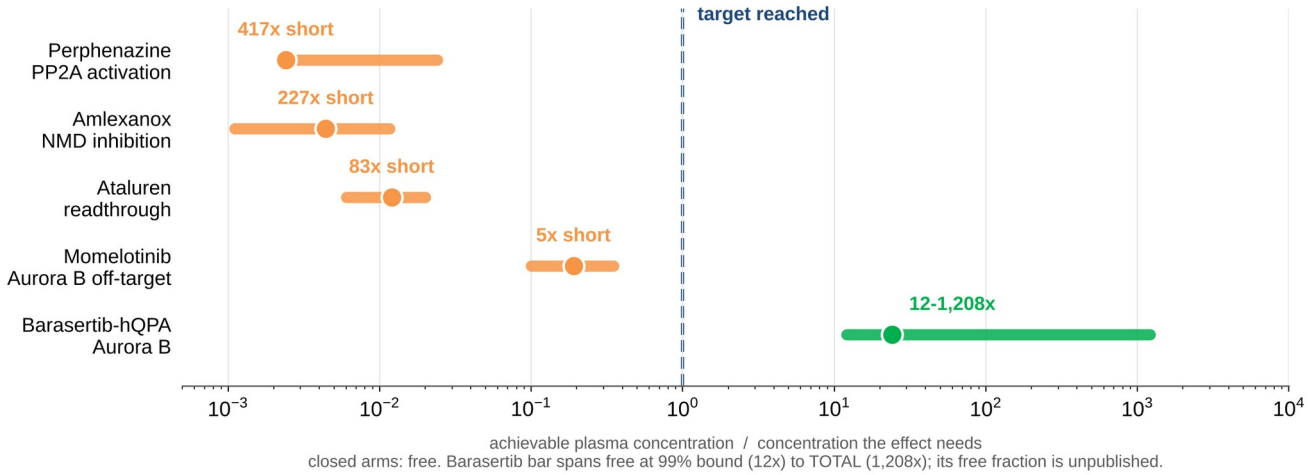
- ~~Direct small-molecule binding~~

closed on exposure

Rebalance downstream
where both alleles meet

- ~~Activate PP2A directly~~
- ~~Restrain APC/C~~
- **Lower Aurora B**

SURVIVES

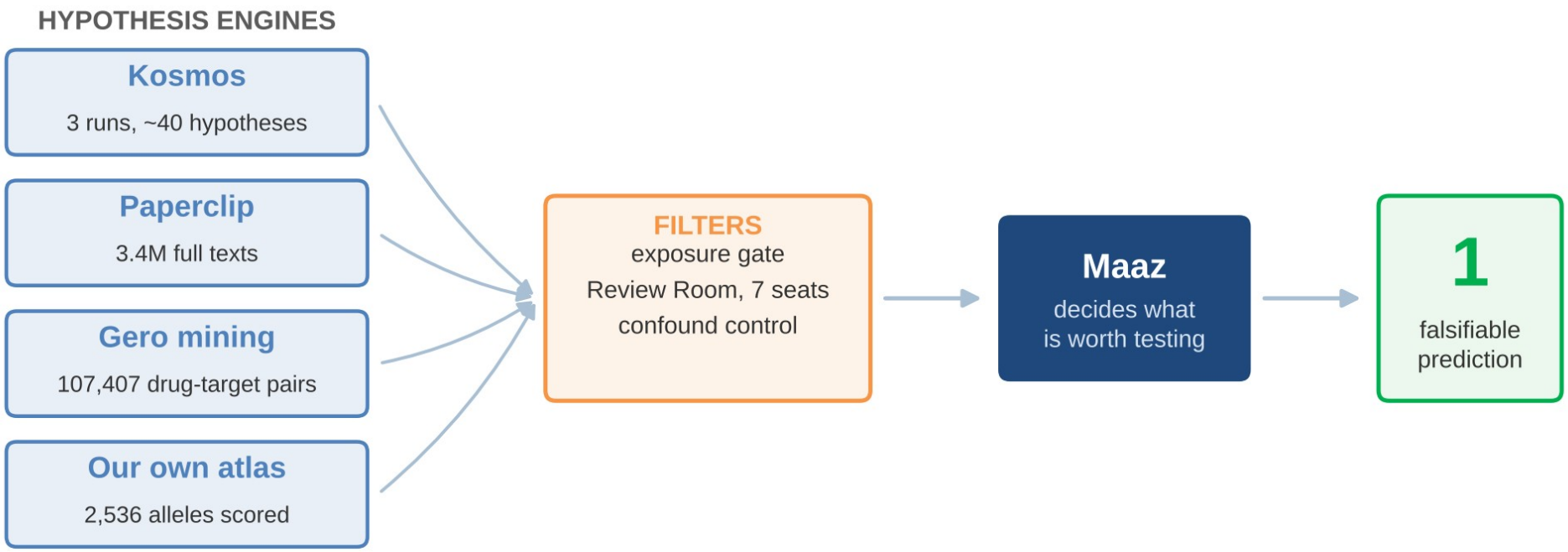


Two gates closed these arms. EXPOSURE: most needed more drug than a person can be given. SAFETY: proteasome and HSP90 inhibitors are carcinogenic, disqualifying where tumour risk comes from missegregation.

Three themes die upstream. One node survives.

INNOVATION (25%) — many engines generate, one human decides, nothing rests on one paper

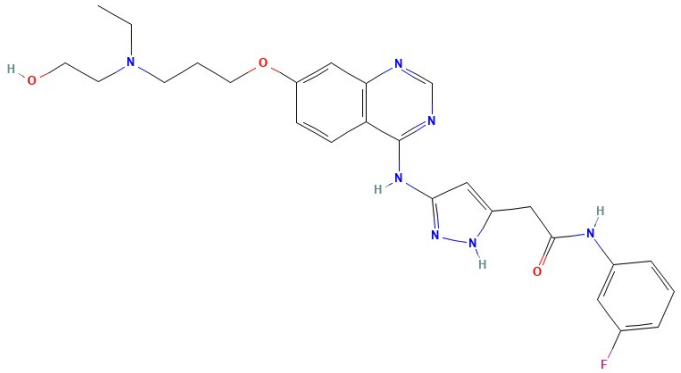
Innovation 25%



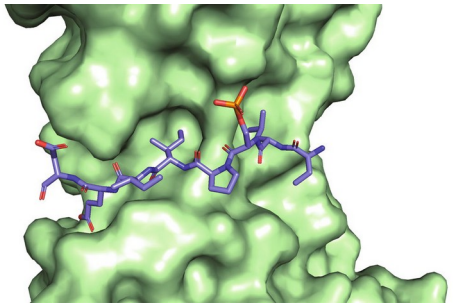
~40 hypotheses → 10 arms survived scrutiny → 1 committed prediction

54	5	4	2,536	51	0
primary papers read, not cited	drug-target databases	experimental structures	alleles scored in our atlas	numbered corrections to our own work	claims resting on a single paper

POTENTIAL IMPACT (25%) — Q4: lower the kinase, because the phosphatase cannot be raised



AZD2811-NP (barasertib-hQPA)



PP2A-B56 with the BubR1 KARD bound — the groove neither allele reaches

OUR PICK AZD2811-NP — AstraZeneca, in Phase II now

Built to do KILL dividing cells — SCLC, AML, lymphoma

Status Phase I/II. Never approved. Not withdrawn for safety.

Dose-limiting tox neutropenia, from ABLATING Aurora B in marrow

Why the NP 2-h infusion not 7-day, and the live asset

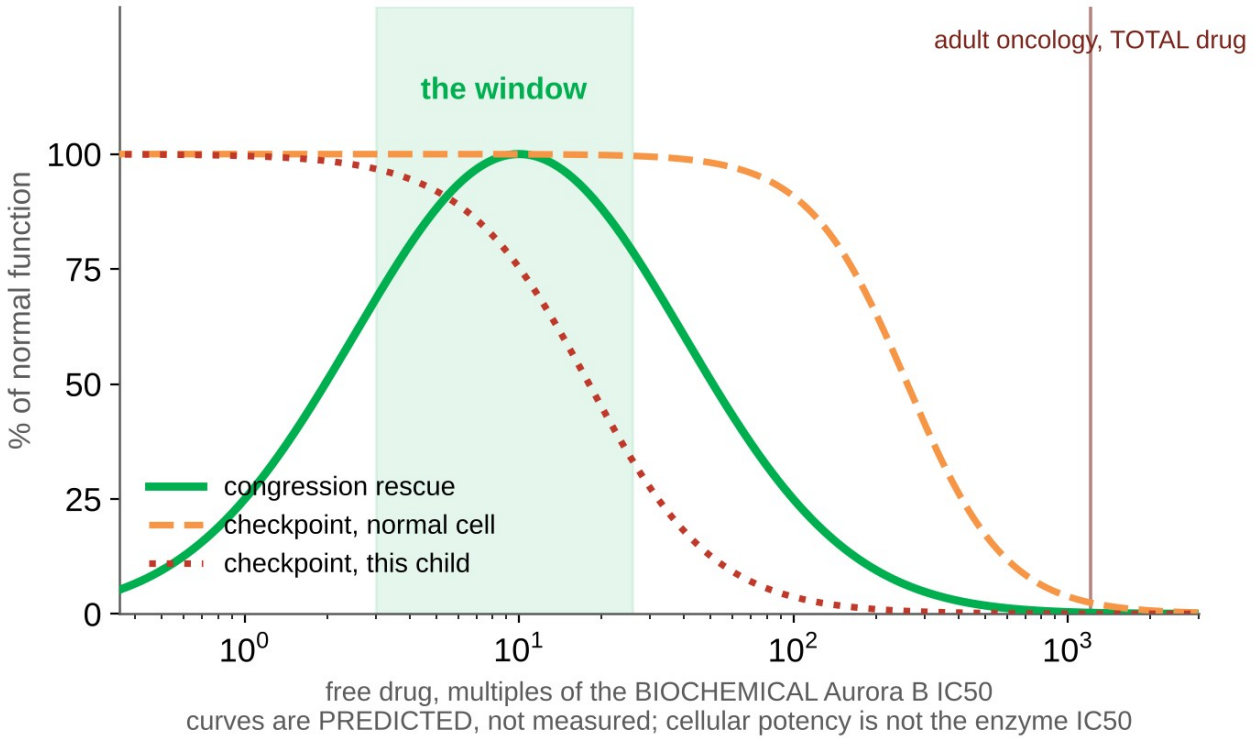
That toxicity is the ARGUMENT, not the obstacle

It comes from ABLATING Aurora B. We want 30-50% engagement.

Alisertib? Oral, given to 139 children — but it targets Aurora A, the wrong kinase. Caveat we own: the NP releases over days.

0.37 nM	**3,700x**	**<100 mg**
Aurora B IC50	over Aurora A	our proposed dose

The dose, anchored on human data: BELOW the dose that first touches marrow



AZD2811 phase I mapped this in humans

51 patients, 15-600 mg. Neutropenia is the ON-TARGET marker of Aurora B engagement — and the toxicity we must avoid. At the oncology dose it is severe. That is the argument for going far below it.

500 mg

oncology RP2D + G-CSF.
15 of 21 had grade 3+ AEs

200 mg

where marrow pharmacology
FIRST appears

≤100 mg

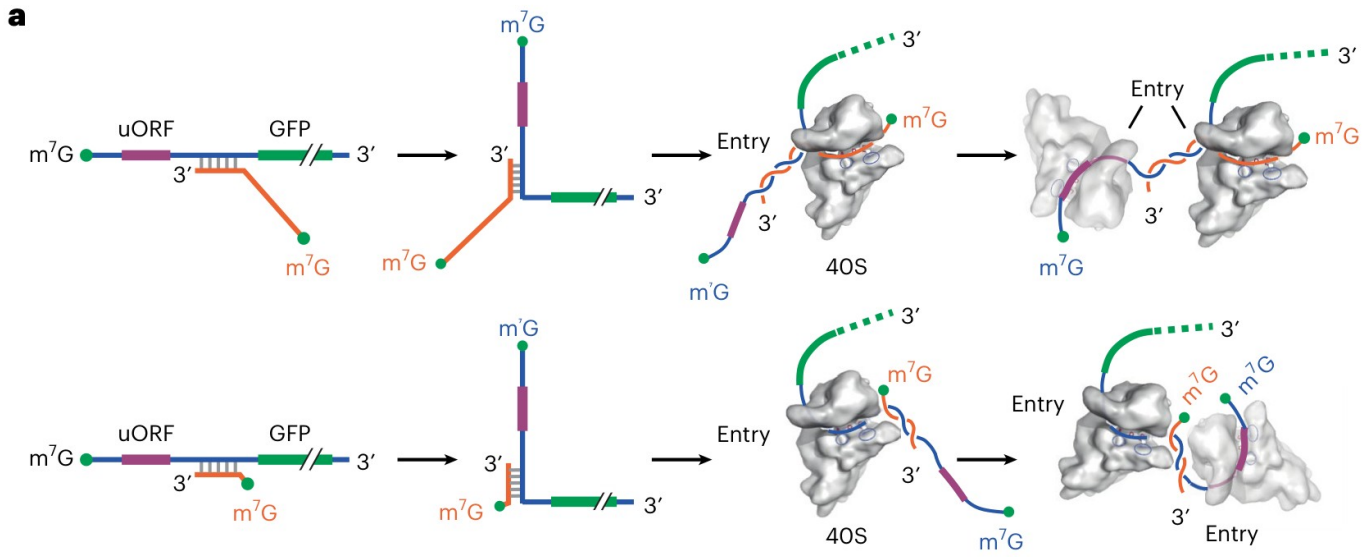
cohorts 1-5: NO neutropenia,
NO dose-limiting toxicity

We propose BELOW 100 mg/cycle

a 5x reduction, in the SAME formulation, titrated on phospho-Hec1

Falchook 2023 PMID 36871042 · Johnson 2026 PMID 41933194

The moonshot: programmable translation, on the message the patient already has



What it does

A separate capped RNA (orange) base-pairs with the target mRNA (blue). The ribosome loads from the trans-cap and is handed to a start codon of our choosing.

Why it reaches this child

Allele 1's transcript is gone, so nothing can rescue it. But allele 2's message IS present — raising its output is a genome-free way to restore the dose.

Why it deserves funding

One mechanism, no edit, and dosage disease is a whole class. Delivery is the unsolved half: liver and muscle today, brain and marrow not yet.

Jia et al., Nat Biotechnol 2025, Fig 1a

What we answered, and the one prediction we commit to

Q1 What do the alleles leave? No transcript, and an unstable protein. Both lose the KARD's function.

Q2 Is it reversible? Yes — restoring BubR1 doubles checkpoint arrest in patient cells.

Q3 Can a drug raise BubR1? No. Zero compounds across 107,407 curated pairs, and no PP2A activator.

Q4 What else sits on the node? Aurora B — the one target where human exposure runs in our favour.

The prediction: partial Aurora B inhibition rescues congression in this child's cells BEFORE it costs checkpoint competence. If it does not, we are wrong.

Maaz Asher Khurram · mk7790@nyu.edu